Author’s Note

During my entire research life, friends and curious people have asked me to explain what was going on in quantum gravity research. How was it possible to study new ways of thinking about space and time? Over and over again I have been asked to write a popular account of this research. While books on cosmology or string theory abound, a book describing the research on the quantum nature of space and time, and on loop quantum gravity in particular, did not yet exist. I have long hesitated, because I wanted to concentrate on research. Some years ago, after completing my technical book on the subject, I felt that the collective work of many scientists had moved the topic to a stage mature enough for a popular book. The landscape we are exploring is enchanting: why keep it hidden?

But I still delayed the project, because I could not ‘see’ the book in my head. How to explain a world without space and time? One night in 2012, during a long solitary drive from Italy to France, I realized that the only way to explain in a comprehensible manner the ongoing modifications of the notions of space and time was to tell the story from the beginning: starting from Democritus, all the way through to the quanta of space. After all, this is how I understand the story. I began to design the entire book in my mind while driving, and got increasingly excited, until I heard a police car’s sirens telling me to pull over: I was driving far above the speed limit. The Italian policemen asked me politely if I was crazy to drive at that speed. I explained that I had just found the idea I’d been seeking for so long; the policeman let me go without a ticket, and wished me good luck with the book. This is the book.

This book was written and first published in Italian at the beginning of 2014. Shortly afterwards, I wrote a few articles on fundamental physics for an Italian newspaper. A prestigious Italian publisher, Adelphi, asked me for an extended version of these articles, to appear as a small booklet. This is the origin of the short book *Seven Brief Lessons on Physics*, which to my immense surprise has become an international bestseller and has opened a beautiful channel of communication between me and so many wonderful readers all over the world. The *Seven Lessons* were thus written *after* this book, and to some extent they are a synthesis of some of the topics you find here. If you have read *Seven Brief Lessons on Physics* and want to know